Strata Cobot: Autonomous Ore Sorting and Haulage with a Manipulator and a Mobile Robot in ROS 2

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***Abstract*—This paper presents a robotics course project in which a UR7e manipulator and a four-wheeled mobile robot (eBot) cooperate to sort and haul ore in a Gazebo simulation. The manipulator detects ores with an RGB-D camera, picks each with a magnetic tool and drops it in the container for its mineral. The mobile robot plans its own collision-free paths on an occupancy grid and follows them while collecting an ore package. All perception, inverse kinematics, motion planning and path planning were written from first principles, without ready-made planners such as MoveIt 2 or Nav2. Ore positions are estimated to within 1–3 mm, inverse kinematics plans in 0.1–0.4 s, and planned paths are as short as a reference shortest path. Across repeated trials the manipulator sorted every ore into the correct container and the mobile robot completed every lap, with no collision and no safety stop.**

***Index Terms*—ROS 2, mobile manipulation, inverse kinematics, A* path planning, pure pursuit, RGB-D perception**

I. Introduction

The project is set in a mining operation on another planet run by two robots with no human in the loop. A mobile robot crosses rocky terrain to an excavation site, takes on ore and hauls it back to a processing facility. There, a stationary robot arm identifies each ore with a camera and sorts it into one of three containers: azurite (blue), malachite (green) or vanadinite (orange). The simulated world, robots and sensors come from the e-Yantra Strata Cobot environment [1].

The project has two parts. The manipulator must find four ores on a conveyor, pick each with an electromagnet and drop it in the container for its mineral. The mobile robot must drive a three-leg lap between fixed poses through a field of rocks, collecting an ore package on the way, along paths it plans itself. Every planner and controller in this work is our own; no motion planning or navigation library is used.

The contributions of this project are:

- a colour and depth pipeline that publishes each ore's 3D position with 1–3 mm error;
- a pick-and-place pipeline built on damped least-squares inverse kinematics and resolved-rate control, with force-confirmed picks;
- an occupancy-grid planner whose paths match the reference shortest length, and a pure-pursuit follower that tracks them within 0.09 m;
- an evaluation of both pipelines over repeated randomised trials.

II. System Overview

The system runs on Ubuntu 24.04 with ROS 2 [6] Jazzy and Gazebo Harmonic. It consists of four Python nodes, two per robot (Fig. 1). The arm accepts only joint or tool velocity commands, and the mobile robot only velocity twists on `/cmd_vel`, so every node closes its own control loop at 20–50 Hz.

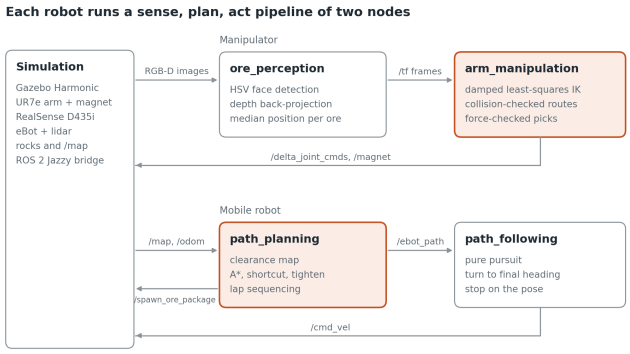


Fig. 1. Node architecture. Feedback topics (`/tcp_pose_raw`, `/joint_states`, the tool force and `/arm_status` for the arm; `/odom` for the mobile robot) are omitted for clarity.

The arm is a UR7e with an electromagnet on its flange and a RealSense D435i camera fixed to its base, looking down at a conveyor. Its servo caps each joint at 0.60 rad/s and triggers a latching protective stop on joint limits, singularities, collisions and overload. The mobile robot is a skid-steer base with a 2D lidar; a static occupancy grid of the arena is published once on `/map`.

III. Perception

Each ore is a 101.6×101.6×76.2 mm block, black except for one coloured top face. The detector converts each colour frame to HSV and thresholds one range per mineral. The faces are far more saturated ($S > 200$) than the orange terrain ($S < 175$), which separates vanadinite from the ground. A blob is kept only if its area is plausible and it fills at least 60% of its minimum-area rectangle, since faces are square.

Each pixel (u, v) of a face is back-projected with its own aligned depth z through the camera intrinsics,

$$\mathbf{p}_c = \left(\frac{(u - c_x)z}{f_x}, \ \frac{(v - c_y)z}{f_y}, \ z \right), \tag{1}$$

and transformed into the arm's `base_link` frame. The median of these points is the face centre, which is robust to noisy depth along the edges.

The number of ores of each mineral varies between trials, so the detector observes 20 frames before naming anything. Detections of one mineral closer than 5 cm are clustered as one ore, clusters seen in fewer than a third of the frames are discarded, and each mineral is numbered from 1, nearest the arm first. Only faces inside the conveyor's volume are accepted, so an ore already on the tool or in a container never moves a published position. Each ore is published as a tf2 frame at 10 Hz, at the median of its last 30 detections (Fig. 2). Compared with the simulator's ground truth, the published positions were 1–3 mm from the true top-face centres.

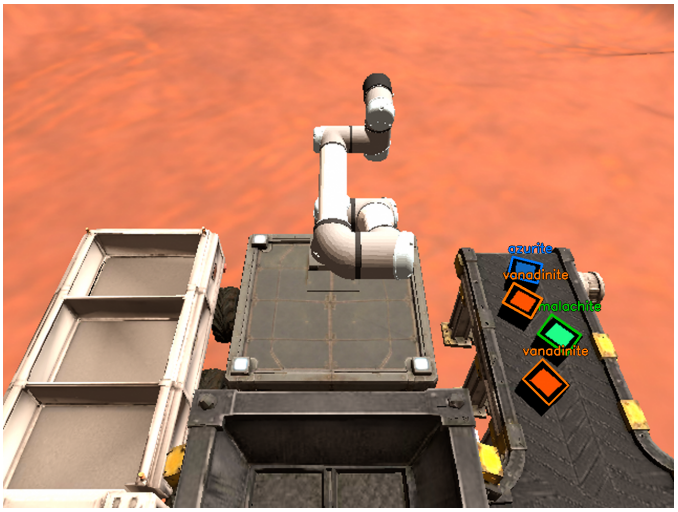


Fig. 2. RealSense colour frame with four detected ores on the conveyor (right). The cargo rover's three containers are at the left.

IV. Manipulation

A. Workspace and containers

The container positions are not given. They were extracted by parsing the cargo rover's collision mesh (a GLB file) and transforming its vertices into `base_link`. Each container's interior spans 0.40 m in y and 0.26–0.31 m in x. The centre of the vanadinite container lies 0.93 m from the arm base, beyond reach with the tool vertical, so each ore is released over the point of its container nearest the arm, 8 cm in from the walls (Fig. 3).

B. Inverse kinematics

The robot model, including a point cloud of each link's collision mesh, is read from `/robot_description`. Because the magnet face is a disc, rotation about the vertical is free, and inverse kinematics solves five constraints: the tool position $\mathbf{p}$ and a vertical tool axis $\mathbf{z}$. With error $\mathbf{e} = [\mathbf{p}^* - \mathbf{p};\ (\mathbf{z} \times -\hat{\mathbf{k}})_{xy}]$ and the corresponding 5×6 rows J of the numerical Jacobian, each iteration applies the damped least-squares step [4]

$$\Delta\mathbf{q} = J^T \left(JJ^T + \lambda^2 I\right)^{-1} \mathbf{e}, \qquad \lambda = 0.02, \tag{2}$$

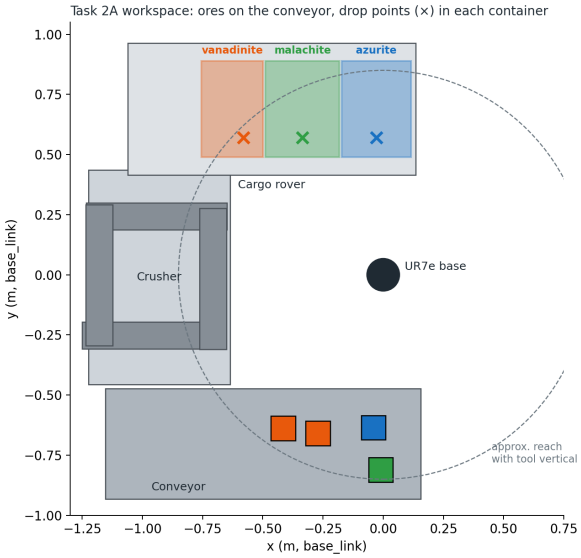


Fig. 3. Top view of the arm workspace in `base_link`: equipment modelled as boxes for collision checking, ores on the conveyor, and the drop point (×) in each container.

clamped to 0.3 rad. A converged solution is accepted only if it lies inside the joint limits with a 0.3 rad margin, keeps the arm 3 cm from every obstacle box, keeps the tool within 1° of vertical and keeps the smallest singular value of the Jacobian above 0.04. Each accepted configuration is stored, and the stored configurations nearest a new target are used as extra seeds. Without these seeds, IK started above the containers failed to find its way back to the conveyor. An earlier version used a constrained SLSQP optimiser and took minutes per plan; the solver of (2) plans in 0.1–0.4 s.

C. Motion and picking

Large moves are straight lines in joint space, accepted if the swept configurations keep 2.5 cm of clearance. Otherwise the planner tries a lift–traverse–descend detour at 0.30–0.50 m, then a route through a staging point over open floor. While an ore is carried, its block is added to the collision geometry hanging below the tool. Joint targets are tracked with a proportional controller capped at 0.30 rad/s.

The magnet face is an 81 mm disc and the ore face 101.6 mm square, so the tool must arrive within about 10 mm of the face centre and parallel to it. Vertical moves therefore use resolved-rate control [5]: a desired tool velocity $\mathbf{v}$, proportional to the position error and capped, and a tilt correction ω are mapped to joint velocities by $\dot{\mathbf{q}} = J^{+}[\mathbf{v}; \omega]$. Each pick hovers 10 cm above the face, switches the magnet on, descends at 4 cm/s and then 1.2 cm/s, and stops at the face or when the contact force exceeds 2.5 N. After lifting, the tool force (tool weight removed) reads about 1.47 N if a 0.15 kg ore is held; below 0.8 N the pick is retried, up to three times. If a protective stop latches, the node unlocks the arm, backs off upward and re-plans the interrupted step.

V. Path Planning and Following

A. Problem

The mobile robot drives from home $(0, 0)$ to the ore drop pose $(6.0, 1.9)$, requests the ore package, continues to the arm pose $(0, 1.9)$ and returns home. A path is valid if it starts and ends exactly on these poses and crosses no occupied cell; the robot must stay within 0.5 m of it and stop within 0.3 m and 0.15 rad of each pose. Path quality is measured by efficiency,

$$\eta = \min\left(1,\ \frac{L_{\text{ref}}}{L_{\text{path}}}\right), \tag{3}$$

where the reference length L_{ref} is the shortest path for a 0.395 m circle enclosing the 0.62×0.49 m chassis.

B. Planner

A Euclidean distance transform of the 5 mm `/map` (1784×1380 cells) gives every cell its clearance to the nearest rock or wall. A 5 cm grid sampled at cell centres is searched by 8-connected A* [2] over the cells whose clearance meets a chosen level c. Near the fixed start and end poses the limit drops to 0.30 m, since the ore drop pose lies only 0.375 m from the arena wall. The A* route is then shortcut, joining each point to the furthest later point visible with clearance c, and tightened: each corner is pulled towards the chord between its neighbours, halving the step until both legs still clear. Every straight leg is verified on the full-resolution map at 5 mm spacing. Tightening removes the grid's quantisation, so the path wraps each rock at exactly the clearance asked for (Fig. 4).

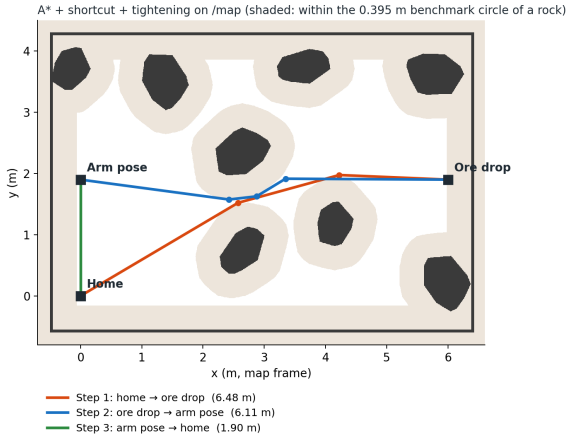


Fig. 4. Planned paths for the three legs of the lap. Shaded cells lie within 0.395 m of a rock or wall.

C. Choosing the clearance

The reference path treats the robot as a 0.395 m circle, but when driving straight past a rock only its 0.245 m half-width must clear. Fig. 5 shows the trade-off on one rock layout: below about 0.40 m the lap length is nearly flat, while above it narrow gaps close and the lap grows by up to 13%. The planner first plans at 0.395 m to estimate the reference path, then tries levels from 0.40 down to 0.34 m and selects the safest whose length is within 1 cm of that estimate. At 0.38 m the robot keeps 13.5 cm to each side, against a measured tracking error of at most 9 cm.

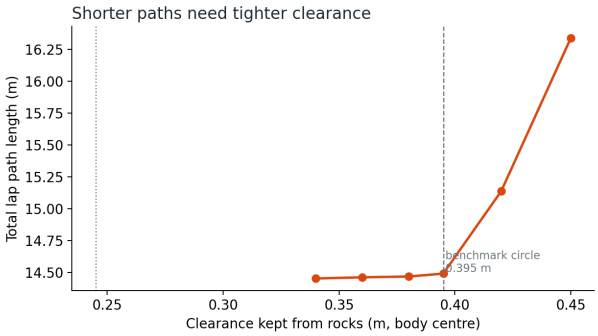


Fig. 5. Total lap length against the clearance kept from rocks, for one rock layout. No path exists above 0.45 m.

D. Follower

The follower is a pure-pursuit controller [3] with a 0.30 m lookahead and a 0.35 m/s cruise speed. It turns on the spot when the heading error exceeds 40°, slows on the approach to the end of the path, then turns to the path's final heading and stops. It drives the published path without re-planning; the lidar only slows it near obstacles. The planner watches `/odom`, advances to the next leg after the robot has stood still on the end pose for 0.5 s, and at the ore drop pose calls `/spawn_ore_package` asynchronously.

VI. Results

Each trial used a freshly launched world with randomised ore positions or rock layout, and was assessed by the automated evaluator supplied with the simulation. Table I summarises the trials.

TABLE I
Trial Results

Trial	Outcome
Arm 1	4/4 ores detected, picked and dropped in the correct container; 0 safety stops
Arm 2	Different ore mix, one ore 0.83 m from the base: 4/4 sorted correctly in 167 s; 0 safety stops
Robot 1	First planner: 14.74 m lap vs. 14.46 m reference, η = 0.98
Robot 2	Final planner: 14.55 m vs. 14.58 m, η = 1.00
Robot 3	Final planner: 14.48 m vs. 14.43 m, η = 0.996

All robot trials collected the ore package without a collision, kept the robot within 0.09 m of its paths and stopped within 0.15 m and 0.04 rad of each pose. Every pick was confirmed at 1.46–1.47 N. The small shortfall in trial Robot 3 comes from the reference itself: on the same rock layout the reference length of the second leg measured 6.14 m in one trial and 6.07 m in the next, while the planned path stayed at 6.11 m. Closing that gap reliably would require clearances below 0.34 m, at a real risk of collision.

VII. Challenges and Lessons

Table II summarises the main problems found during testing. Most were invisible until the robots were run end to end and assessed automatically.

TABLE II
Main Problems and Fixes

Problem	Cause and fix
No path found	Drop pose 0.375 m from the wall, plus clearance lost to map downsampling. Fixed by full-resolution clearance and a relaxed limit near fixed poses.
Paths 2% long	Coarse-grid checks and grid-snapped corners. Fixed by fine-map checks, tightening and reference-aware clearance.
IK slow or stuck	SLSQP evaluated all checks in every gradient step; poor seeds over the containers. Fixed by DLS IK and a seed library.
Vanadinite unreachable	Container centre 0.93 m away. Fixed by releasing at the nearest reachable point inside it.
Ores never found	tf frame list writes a space after each name. Fixed by a regular-expression parse.

Three lessons stand out. Testing end to end against an automatic evaluator early exposed losses that plots did not. Measuring the geometry ourselves revealed both the wall beside the drop pose and the unreachable container centre. And a solver specific to the problem, with checks applied once to its result, was about 300 times faster than a general constrained optimiser.

VIII. Conclusion

Both robots complete their missions end to end in simulation, using only planners and controllers written for this project: every ore was sorted correctly and every lap was completed without a collision. Future work includes a handoff in which the mobile robot delivers ore to the arm, re-planning around obstacles the map does not contain, faster sorting by blending moves and ordering picks, and a move to physical hardware, where lighting and sensor noise will test the colour thresholds and the force-based pick check.

Acknowledgment

We thank e-Yantra, IIT Bombay, for the Strata Cobot simulation environment and its evaluation tools.

References

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